designer itself.

4. Both the image sprites will be moving together; hence, they will also have the same set of properties.
5. Next is another horizontal arrangement, which has two buttons with the width set to fill the parent and the height as automatic. Set the text for these buttons accordingly.
6. Then comes the clock component. For the clock, uncheck the enabled property and set the interval as 1 millisecond (1/1000th of a second).
7. For the notifier, keep the default settings as available in the designer.

Now, we move on to the blocks editor to define the behaviour so let's discuss the actual functionality that we expect from the application.

1. We need to code it so that on pressing the *Left* button the paddles should move towards the left, and vice versa.
2. The ball should move independently over the canvas and on touching the walls or paddles, it should bounce back.
3. Based on the game's progress, our score should be increased.
4. We need to keep monitoring the position of the ball, so that if it goes beyond the paddles, we should decrease one point.
5. We should define the basis on which the game ends.
6. As the game finishes, we should display a pop-up enabling the user to choose either to 'Restart' or 'Quit'.

So let's move on and add these behavioural aspects using the block editor. I hope you remember how to switch from the designer to the block editor. There is a button available right above the *Properties* pane to switch between the designer and block editor.

Property	Value
Enabled	FALSE
Heading	0
Interval	100
Paint Colour	Orange
Radius	8
Speed	10
Visible	TRUE
X	171
Y	330
Z	1

Figure 3: Ball properties

Block editor blocks

I have already prepared the blocks for you. All you need to do is drag the relevant blocks from the left side palette and drop them onto the viewer. Arrange the blocks in the same way as shown in the images ahead. I will explain what each does and what they are called.

1. The blocks given in Figures 5 to 9 are easy to understand. The very first one (Figure 5) tells us that when the screen is initialised, i.e., the application has started, it should enable the ball, which will head anywhere between 20 to 70 and enable the clock as well.
2. Now, by touching the buttons, the paddles move left or right. Here, a touchdown refers to how long the button is kept pressed. 'Heading 0' means movement towards the left and '180' indicates movement towards the right. If speed is 10, it means it will move 10 pixels per cycle or interval.
3. Once the buttons are released, the movement of paddles is stopped; hence, speed is once again set at 0 for the touchup blocks.
4. Now if the ball collides with the side walls, it has to be sent back. For this we have used the blocks visible in image 3 (Figure 7) for the block editor. As soon as it collides with the wall, it will bounce back. If the collision is with the upper or lower paddles, then it should move back in the opposite direction; hence, we are using a random block to give it a random heading.
5. The most complex is the fourth block which is for the clock. It says, 'When clock.timer', which means that when the clock is enabled, it will increase the score and will also check for the position of the ball.
6. Once it detects that the ball has moved outside the boundaries, it will decrease the value of the 'points' label we have set in the designer.
7. When the value of the points is less than or equal to zero, it will display the notifier, along with two options, namely 'Restart' and 'Quit'.
8. The last block is to decide what the user wants to do after the notification. If he or she chooses to restart the game, then the score and points will be reset to their default values, and the clock and ball will be enabled once again. If the 'Quit' button is chosen, then it should close the application.

Property	Value
Enabled	FALSE
Heading	0
Interval	100
Picture	Paddle.jpg
Rotates	FALSE
Speed	10
Visible	TRUE
X	100
Y	10 for upper & 345 for lower
Z	1
Width	150
Height	Automatic

Figure 4: Image sprite properties

Now, you are done with the block editor too. Next, we will move to downloading and installing the app on your phone to check how it is working.

Packaging and testing

To test the app, you need to get it on your phone. First, you have to download the application to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, click on the 'Build' button. It will show you the option to download the *apk* to your computer.